

Aim: Write a program implementing basic arithmetic operations of a simple calculator.

Program

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical"
    android:padding="20dp"
    tools:context=".MainActivity">

    <EditText
        android:id="@+id/number1"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Enter Number 01"
        android:inputType="numberDecimal"/>

    <EditText
        android:id="@+id/number2"
        android:layout_width="match_parent"
```

```
android:layout_height="wrap_content"
android:layout_marginTop="10dp"
android:hint="Enter Number 02"
android:inputType="numberDecimal"/>
```

<TextView

```
android:id="@+id/result_text"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:layout_marginTop="20dp"
android:textColor="@color/black"
android:textSize="17sp"
android:textStyle="bold"/>
```

<Button

```
android:id="@+id/add_btn"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:layout_marginTop="10dp"
android:text="+"/>
```

<Button

```
android:id="@+id/sub_btn"
android:layout_width="match_parent"
android:layout_height="wrap_content"
android:layout_marginTop="10dp"
android:text="-"/>
```

<Button

```
android:id="@+id/mul_btn"  
android:layout_width="match_parent"  
android:layout_height="wrap_content"  
android:layout_marginTop="10dp"  
android:text="x"/>
```

<Button

```
android:id="@+id/div_btn"  
android:layout_width="match_parent"  
android:layout_height="wrap_content"  
android:layout_marginTop="10dp"  
android:text="/" />
```

<Button

```
android:id="@+id/clear_btn"  
android:layout_width="match_parent"  
android:layout_height="wrap_content"  
android:layout_marginTop="10dp"  
android:text="clear"/>
```

</LinearLayout>

MainActivity.java

```
package com.example.calculator;
```

```
import androidx.appcompat.app.AppCompatActivity;
```

```
import android.os.Bundle;
```

```
import android.view.View;
```

```
import android.widget.Button;
```

```
import android.widget.EditText;
```

```
import android.widget.TextView;
```

```
import android.widget.Toast;
```

```
public class MainActivity extends AppCompatActivity {
```

```
    EditText number1, number2;
```

```
    TextView result_text;
```

```
    Button add_btn, sub_btn, mul_btn, div_btn, clear_btn;
```

```
    @Override
```

```
    protected void onCreate(Bundle savedInstanceState) {
```

```
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_main);
```

```
        number1 = findViewById(R.id.number1);
```

```
        number2 = findViewById(R.id.number2);
```

```
        result_text = findViewById(R.id.result_text);
```

```
        add_btn = findViewById(R.id.add_btn);
```

```
        sub_btn = findViewById(R.id.sub_btn);
```

```
        mul_btn = findViewById(R.id.mul_btn);
```

```
        div_btn = findViewById(R.id.div_btn);
```

```
        clear_btn = findViewById(R.id.clear_btn);
```

```
        add_btn.setOnClickListener(new View.OnClickListener() {
```

```
            @Override
```

```
            public void onClick(View view) {
```

```
                String number1_text = number1.getText().toString();
```

```
                String number2_text = number2.getText().toString();
```

```
                int num1 = Integer.parseInt(number1_text);
```

```
                int num2 = Integer.parseInt(number2_text);
```

```
float sum
    = num1 + num2;
result_text.setText("The addition of the two numbers is:" + sum);
Toast.makeText(getApplicationContext(), "The addition of the two numbers is:" + sum,
Toast.LENGTH_SHORT).show();
}
});
```

```
sub_btn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        String number1_text = number1.getText().toString();
        String number2_text = number2.getText().toString();
        int num1 = Integer.parseInt(number1_text);
        int num2 = Integer.parseInt(number2_text);
        float sub = num1 - num2;
        result_text.setText("The subtraction of the two numbers is:" + sub);
        Toast.makeText(getApplicationContext(), "The subtraction of the two numbers is:" +
sub, Toast.LENGTH_SHORT).show();
    }
});
```

```
mul_btn.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View view) {
        String number1_text = number1.getText().toString();
        String number2_text = number2.getText().toString();
        int num1 = Integer.parseInt(number1_text);
        int num2 = Integer.parseInt(number2_text);
        float mul = num1 * num2;
```

```
result_text.setText("The multiplication of the two numbers is:" + mul);
```

```
    Toast.makeText(getApplicationContext(), "The multiplication of the two numbers is:" +  
mul, Toast.LENGTH_SHORT).show();
```

```
}
```

```
});
```

```
div_btn.setOnClickListener(new View.OnClickListener() {
```

```
    @Override
```

```
    public void onClick(View view) {
```

```
        String number1_text = number1.getText().toString();
```

```
        String number2_text = number2.getText().toString();
```

```
        int num1 = Integer.parseInt(number1_text);
```

```
        int num2 = Integer.parseInt(number2_text);
```

```
        float div = num1 / num2;
```

```
        result_text.setText("The division of the two numbers is:" + div);
```

```
        Toast.makeText(getApplicationContext(), "The division of the two numbers is:" + div,  
Toast.LENGTH_SHORT).show();
```

```
}
```

```
});
```

```
clear_btn.setOnClickListener(new View.OnClickListener() {
```

```
    @Override
```

```
    public void onClick(View view) {
```

```
        number1.setText("");
```

```
        number2.setText("");
```

```
        result_text.setText("");
```

```
        Toast.makeText(getApplicationContext(), "Inputs cleared...",  
        Toast.LENGTH_SHORT).show();  
    }  
});  
}  
}
```